

Module 1: Network + Network Fundamentals and building networks

Day 4

Task 1:

Dividing 192.168.10.0/24 into 4 equal subnets
/24 -> /26

Subnet	Network	Broadcast	Usable IP
Subnet 1	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
Subnet 2	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126
Subnet 3	192.168.10.128	192.168.10.191	192.168.10.129 - 192.168.10.190
Subnet 4	192.168.10.192	192.168.10.225	192.168.10.193 - 192.168.10.224

Task 2:

Device	IP address
Laptop	192.168.10.10
Phone	192.168.10.11
Smart TV	192.168.10.12
Printer	192.168.10.13
Router (gateway)	192.168.10.14

Task 3:

Subnet 1 - /26

192.168.10.1 - 10.62

(laptop - 192.168.10.10 ,
Phone - 192.168.10.11,
Smart TV - 192.168.10.1)

Subnet 2 - /26

192.168.10.65 - 10.126

Subnet 3 - /26

192.168.10.129 - 10.190

Subnet 4 - /26

192.168.10.193 - 10.254

Task 4:

Step-by-step explanation (as if for a friend new to subnetting)

Step 1 — Figure out how many subnets you need.

You want 4 subnets. Ask: "what power of 2 gives me 4?" Answer: $2^2 = 4$. So you need to borrow **2 extra bits** from the host portion of the address.

Step 2 — Work out the new subnet mask.

Your original network is /24, meaning 24 bits are already used for the network part. Borrowing 2 more bits makes it **/24 + 2 = /26**. In dotted-decimal, /26 converts to **255.255.255.192** (since the last octet becomes 11000000 in binary, which is $128+64=192$).

Step 3 — Find the "block size."

The block size tells you how far apart each subnet's starting address is. It's calculated as 256 minus the last non-255 octet of the mask: $256 - 192 = 64$. So each subnet is 64 addresses wide, and they'll start at 0, 64, 128, 192.

Step 4 — List out each subnet's range.

Starting from 192.168.10.0 and adding 64 each time:

- 192.168.10.**0** to .63
- 192.168.10.**64** to .127
- 192.168.10.**128** to .191
- 192.168.10.**192** to .255

The first address in each range is the **network address** (can't be assigned to a device), and the last is the **broadcast address** (also can't be assigned).

Step 5 — Calculate usable hosts per subnet.

With /26, you have $32 - 26 = 6$ host bits left. That gives $2^6 = 64$ total addresses, minus 2 (network + broadcast) = **62 usable addresses** per subnet.

Step 6 — Assign IPs to real devices..